

Original Reports

Conditioned Placebo Analgesia Persists When Subjects Know They Are Receiving a Placebo

Scott M. Schafer,^{*} Luana Colloca,[†] and Tor D. Wager^{*}

^{*}Department of Psychology and Neuroscience, University of Colorado Boulder, Boulder, Colorado.

[†]School of Nursing, Pain and Translational Symptom Science, University of Maryland Baltimore Center to Advance Chronic Pain Research; and School of Medicine, Department of Anesthesiology, University of Maryland Baltimore, Baltimore, Maryland.

Abstract: Belief in the effectiveness of a placebo treatment is widely thought to be critical for placebo analgesia. Many types of placebo responses—even those that depend on conditioning—appear to be mediated by expectations that are strengthened as treatment cues are reinforced with positive outcomes. However, placebo effects may occur even when participants are aware they are receiving a placebo. To address the question of whether conditioned placebo analgesia can persist in the absence of expectations, we studied the effects of long (4 days) versus short (1 day) conditioning to a placebo treatment. After an initial placebo test, a “reveal” manipulation convincingly demonstrated to participants that they had never received an active drug. Placebo analgesia persisted after the reveal in the long conditioning group only. These findings suggest that reinforcing treatment cues with positive outcomes can create placebo effects that are independent of reported expectations for pain relief.

Perspective: This article demonstrates a form of placebo analgesia that relies on prior conditioning rather than current expected pain relief. This highlights the importance of prior experience on pain relief and offers insight into the variability of placebo effects across individuals.

© 2015 by the American Pain Society

Key words: Placebo, pain, conditioning, expectancy, reversal.

Placebo analgesia is pain relief observed following administration of a treatment that is not directly caused by pharmacological properties of that treatment. Placebo analgesia is typically induced in the laboratory using a “response conditioning” paradigm, where treatment cues (eg, a cream or injection) are paired with

surreptitious reductions in the intensity of painful stimuli.^{25,33} Afterward, painful stimuli are presented under placebo (paired) and control (unpaired) conditions to test for placebo effects. This procedure is a model paradigm in the study of placebo analgesia and the influence of expectations on pain and other affective, perceptual, and physiological processes.^{24,31,35}

Early studies concluded that the experience of pain relief was critical for reliably inducing placebo analgesia,^{33,34} but it is now generally understood that placebo analgesia is directly mediated by expectations and only indirectly relies on prior experiences.^{2,6,19,21,24} Manipulations of expectations produce pain relief,^{2,7} and greater expectancies are associated with greater placebo analgesia.^{18,21,22,25,37} Even within conditioning paradigms, expectancies appear to be critical: When subjects attribute pain relief to sources other than a placebo treatment, they do not acquire placebo analgesia,^{21,38} and verbal suggestions of hyperalgesia can block conditioned placebo analgesic effects.^{6,7,14}

Received September 18, 2014; Revised December 3, 2014; Accepted December 29, 2014.

Research reported in this publication was supported by the National Institute of Mental Health of the National Institutes of Health under Award Number R01MH076136, by the National Institutes of Health under Grant Number 5F31DA034516-03 and by the Intramural Program of the National Institute of Mental Health under Grant Number Z99 AT999999. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. The authors declare no conflicts of interest.

Supplementary data accompanying this article are available online at www.jpain.org and www.sciencedirect.com.

Address reprint requests to Scott M. Schafer, MA, Department of Psychology and Neuroscience, University of Colorado Boulder, 345 UCB, Boulder, CO 80309-0345. E-mail: scott.schafer@colorado.edu
1526-5900/\$36.00

© 2015 by the American Pain Society

<http://dx.doi.org/10.1016/j.jpain.2014.12.008>

These findings fit within a broader literature suggesting that conditioning depends on the information value of cues rather than associative pairing per se²⁶ and may reflect inferential rather than gradual learning processes.¹²

Expectancy theory implies that belief in the placebo is critical for placebo analgesia. This expectation need not be a belief in the chemical analgesic properties of the treatment but may instead be a more general belief that a placebo treatment can relieve symptoms. This belief may allow placebos to serve as either dose extenders for chemically active treatments^{28,29} or effective treatments on their own.¹⁷ However, expectancy theory is challenged by demonstrations that placebo treatments can result in analgesia even when participants are unaware they are receiving a treatment.^{2,15} Other placebo manipulations that generate expectancy-independent placebo effects (eg, conditioned immunosuppression) generally use multiple conditioning sessions,^{1,6} and increasing the number of conditioning sessions leads to placebo analgesia that is both stronger and more resistant to extinction.¹⁰ A key question is whether enhanced placebo analgesia following multiple conditioning sessions also depends on expectancy. If not, this suggests the existence of a class of placebo analgesia that depends on conditioned associations³ and, like conditioned immunosuppression, is independent of expectations. These placebo effects should depend on the duration of conditioning, be independent of reported expectations, and persist when expectations are reversed.

In order to determine whether conditioned placebo analgesia persists when subjects are made aware of a placebo treatment, pain response was tested both before and after a complete and convincing disclosure of the placebo manipulation (placebo reveal). To directly measure the role of associative learning in "open-label" placebo effects, we varied the number of conditioning sessions and tested whether postreveal placebo effects were greater for participants who had experienced more conditioning sessions. Critically, we measured expected pain relief both before and after the placebo reveal, as nonconscious cues may continue to elicit expectations for pain relief.¹⁵ We hypothesized that participants who experienced more conditioning would engage mechanisms for placebo analgesia that were independent of reported

expectancies and would continue to show placebo analgesia even when aware that the treatment was a placebo.

Methods

Participants

Fifty-four participants (30 female, ages 18–55) were recruited via online advertisements on a recruitment website managed by the School of Medicine at the University of Colorado Anschutz Medical Campus. Data collection was planned to continue until 40 participants met inclusion criteria and completed the study. Twelve participants were excluded during an initial calibration because they did not find the thermal stimuli sufficiently painful (average pain rating below 30 on a 100-point visual analog scale [VAS] for a 48°C stimulus), and 2 participants stopped participation midway through the study because of discomfort from the heat. It was also required that participants' pain ratings increase with higher stimulation temperatures during the initial calibration ($R^2 > .40$), but no participants were excluded on the basis of low temperature discriminability. A total of 40 participants were included in the final analysis, 20 in the long conditioning group (long; 13 female participants) and 20 in the short conditioning group (short; 14 female participants). All participants gave informed consent to participate in a study of treatment effects on pain relief and were fully debriefed at the conclusion of the study. This study was approved by the University of Colorado Boulder Institutional Review Board.

Materials and Procedures

Overview

Participants were informed that they were participating in a study to compare the analgesic effects of a topical cream containing an active analgesic component (placebo cream) to those of a topical cream containing no active ingredients (control cream). Following the initial calibration phase, subjects were randomized to long or short conditioning groups and began the conditioning phase of the study. Immediately following

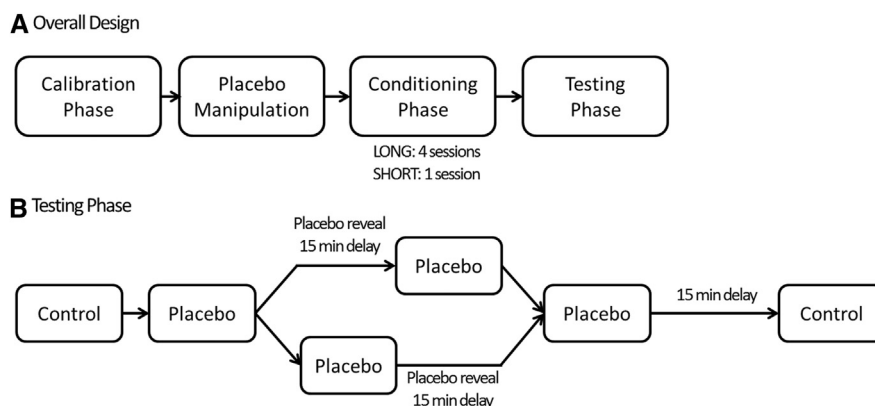


Figure 1. Study design. (A) Participants in the long group had 4 sessions during the conditioning phase and participants in the short group had a single session. (B) During the testing phase, the placebo reveal occurred after the first placebo run for half of all subjects and after the second placebo run for the remaining subjects.

the conditioning phase, placebo analgesia was measured during the testing phase both before and after subjects were told the treatment was a placebo (placebo reveal) (Fig 1). Placebo analgesia was measured as the difference in reported pain between placebo and control stimulations at identical temperatures. All thermal pain stimulations were delivered from a 16×16 mm thermode (Medoc, Ltd, Ramat Yishai, Israel) and lasted for 20 seconds at peak temperature.³²

Calibration Phase

During the initial calibration, participants received 16 thermal stimulations on their left forearm at 8 different sites. Each site received 1 high-temperature stimulus (45, 46, 47, or 48°C) and 1 low-temperature stimulus (41, 42, 43, or 44°C). During stimulation, participants were asked to continuously report how much pain they were experiencing on a 100-point VAS, where 0 was "no pain experienced" and 100 was "the most pain imaginable." These continuous pain ratings were averaged within each trial to create a single pain value (equivalent to an area-under-the-curve measure up to scaling) associated with each stimulation. The overall ratings were regressed onto temperature, and the 4 sites with the lowest residual errors were used for the remainder of the experiment. The regression was used to derive 6 temperatures for each participant to be used in the remainder of the experiment: 2 low (ratings from 10 to 20), 2 medium (ratings from 30 to 40), and 2 high (ratings from 50 to 60). This difference in pain level between the low and high stimulations has been shown to elicit strong placebo effects.²⁴

Placebo Manipulation

Two creams were used in the study: a control cream and a placebo cream. Both creams were an identical petroleum-based jelly; the only difference between them was the addition of blue food coloring to the placebo cream. Participants were told that the placebo cream contained an active analgesic and were instructed on the nature of the analgesic, including its use, warnings, and potential side effects. Following each application of either cream, participants reported whether they were experiencing any side effects (eg, drowsiness, swelling, labored breathing) as a result of the cream. During debriefing, all participants indicated that they had believed that the placebo cream contained an active analgesic prior to the experimenter's revealing otherwise.

Conditioning Phase

During conditioning, experience with the placebo treatment was manipulated by adjusting how many conditioning sessions each participant completed. Subjects in the long group participated in 4 separate conditioning sessions, whereas subjects in the short group participated in only a single conditioning session. Conditioning sessions given to participants in the long group were each given on separate days, with a maximum of 7 days of separation between sessions

(mean intervening time = 2.42 ± 1.0 days). During each conditioning session, placebo and control creams were administered blockwise, with the order of the blocks counterbalanced across participants such that half of participants were first presented with control cream blocks during conditioning (control first) whereas the other half were first presented with placebo cream blocks (placebo first), counterbalanced with conditioning group. Participants were fully aware of which cream they had received at all times. Following each cream application, participants rated their expectancies for pain relief from that cream using a 0 to 100 VAS, where 0 was "no pain relief" and 100 was "the most pain relief imaginable." After a 5-minute waiting period, the cream was cleaned off of the arm and the thermal stimulation runs were initiated. Each run consisted of 2 stimulations on each of 4 sites for a total of 8 trials per run. Placebo stimulations used the 2 low temperatures, and control stimulations used the 2 high temperatures. Conditioning sessions on days 1 to 3 for the long group contained 2 runs within each cream block (16 trials total per cream). However, on the final conditioning day (day 4 for long, day 1 for short), there was only a single run of 8 stimulations in each cream block. This was designed to reduce the likelihood of habituation during the subsequent testing phase while still providing enough conditioning to develop placebo analgesia in the short group.¹⁰

Testing Phase

The testing phase began 15 minutes after the end of the conditioning phase and involved 5 runs of 8 medium-temperature stimulations each. The first and last runs used the control cream (runs 1 and 5), whereas the middle 3 runs used the placebo cream (runs 2–4). Previously learned beliefs, such as those learned during the conditioning phase, tend to persist even when subjects are given subsequent information that those beliefs may be inaccurate,^{27,30} so before each run, participants were asked to rate how much pain relief they expected from the cream using the same 0- to 100-point VAS discussed previously. Midway through the placebo runs, the true nature of the treatment was revealed, and subjects were informed that the placebo treatment was not a pain-relieving cream. In order to ensure that participants truly believed that both treatment creams were inert, the placebo reveal incorporated both demonstration and verbal information, both of which were designed to lead subjects to attribute their previous pain relief to another source. Specifically, participants were told that 1) both creams were identical with the exception of blue food coloring, 2) the stimulation temperatures during conditioning had been lowered for the placebo cream, and 3) neither cream possessed active analgesic ingredients. Following the reveal, a 15-minute waiting period was imposed before resuming the experiment. During this time, the experimenter demonstrated how the placebo cream was made from the control cream to encourage belief in the reveal. Control cream was removed from the canister, mixed with blue food coloring, and then placed into the placebo cream canister. All subjects reported

being convinced by this demonstration during debriefing. The placebo reveal and subsequent delay occurred following either the first or second placebo run (counterbalanced, overall testing run 2 or 3). There was an additional 15-minute delay between the fourth and fifth testing runs designed to reduce any potential carryover analgesic effects from placebo to control blocks (Fig 1B).

Although the manipulation of belief in the placebo cannot be easily counterbalanced, several aspects of this design mitigate the issue of convolving block order with the reveal.^{2,16,23} Using the control cream during the first and final blocks allows the test of whether there is a habituation effect over the entirety of the testing period. If there is no habituation effect, this provides confidence that differences in pain reports following the reveal are not due to simple habituation mechanisms. Additionally, we controlled for any extinction differences in placebo pain reports due to simple repetition of thermal stimulations by adjusting the timing of the reveal between participants. The second placebo run occurs prereveal for half of the participants and postreveal for the other half. Comparing pain reports during the second placebo run between subjects before and after the reveal allows testing of whether the instructions were effective at increasing placebo pain reports, controlling for the total number of thermal stimulations.

Analysis

Differences in expected analgesia (Placebo – Control) were analyzed using a hierarchical mixed-effects generalized linear model (GLM) over reveal state (Prereveal – Postreveal) and conditioning group (Long – Short). In addition to the hierarchical GLM, we performed 2 planned, independent-samples *t*-tests. The first was used to evaluate differences in prereveal expected analgesia by conditioning group, and the second was used to evaluate differences in postreveal expected analgesia by conditioning group.

Placebo analgesia, calculated as (Placebo – Control) differences in pain, was analyzed using a mixed-effects GLM: a 2-within (Placebo – Control) $\times$ (Postreveal – Prereveal) by 2-between (Long – Short) $\times$ (Control first – Placebo first) model, with temperature (2 medium-intensity values that varied by subject according to their prior calibration) included as a covariate of no interest. Planned contrasts included testing placebo analgesia both prereveal and postreveal, the interaction of analgesia with reveal state, and the interaction of all of the above with conditioning group. Given that these contrasts are not orthogonal to each other, we ran 2 variants of the pain GLM. The first variant was designed to implement a standard analysis of variance design, estimating main effects for each factor and all interactions between them. The second variant was identical to the first, save that prereveal and postreveal analgesia were estimated separately in lieu of the (Placebo – Control) main effect and (Placebo – Control) $\times$ (Postreveal – Prereveal)

interaction, providing planned comparisons of placebo effects prereveal and postreveal.

We also implemented tests in the hierarchical GLM for short and long conditioning groups separately and control first and placebo first groups separately. Conditioning order effects (Control first – Placebo first) were not part of our planned contrasts and were originally included in the model to control for order effects. However, conditioning order was strongly predictive of subsequent placebo analgesia, and so we also report results from the (Placebo – Control) $\times$ (Postreveal – Prereveal) $\times$ (Short vs Long conditioning) model in the control first group, which showed the strongest placebo analgesia.

We ran an additional mixed-effects GLM to test for differences in placebo pain prereveal and postreveal, controlling for the total number of stimulus presentations. This model was a 3-between (Long – Short) $\times$ (Control first – Placebo first) $\times$ (Postreveal – Prereveal) GLM. This model only used pain reports collected from the third testing run, where half of the participants were in the prereveal phase and the other half were in the postreveal phase. The planned contrasts for this model were whether placebo pain was different prereveal compared to postreveal and whether that difference varied by conditioning group (Long – Short). All mixed-effects models of pain were fit using the lme4 package in R (R Foundation for Statistical Computing, Vienna, Austria), allowing the within-subject intercepts and slopes to vary as random effects.⁴ When reporting statistics for mixed-effects models, we report *F* statistics using the most conservative estimates of degrees of freedom.

Finally, we tested whether placebo analgesia was correlated with expected analgesia both prereveal and postreveal and whether that correlation was different between conditioning groups (long vs short).

Results

Habituation

During the testing phase, pain ratings for control stimulations did not change following the placebo reveal ($F_{1,38} = .04$, $P = .84$), and this effect was not different between conditioning groups ($F_{1,38} = .16$, $P = .70$). These results suggest an absence of overall habituation or sensitization effects across time during the testing phase.

Expectancy

VAS ratings of expected analgesia (Placebo – Control) were higher for the placebo cream compared to the control cream prereveal (difference scores of 53.1 ± 13.2 and 39.7 ± 10.7 out of 100 points for long and short groups, respectively) and were not different between conditioning groups ($t_{38} = 1.54$, $P = .13$). Differences in expected analgesia dropped 41 points on average to near-zero levels (6.0 ± 5.7 and 4.7 ± 3.9 for long and short groups, respectively) following the reveal ($F_{1,38} = 81.27$, $P < .001$), and this decrease was not

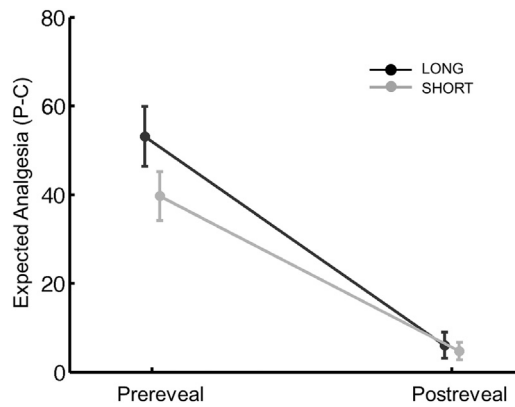


Figure 2. Expectancy. Expected analgesia (Placebo – Control) significantly decreased following the reveal in both the long and short conditioning groups, with no significant differences between groups. The solid dark-gray and light-gray lines represent the long and short groups, respectively. Data are presented as means $\pm$ standard errors of the mean.

significantly different between conditioning groups ($F_{1,38} = 1.79$, $P = .19$). Postreveal expected analgesia was not significantly different between conditioning groups ($t_{38} = .36$, $P = .72$) (Fig 2). The reduction in expected analgesia following the reveal was primarily driven by changes in expectancy for the placebo cream ($F_{1,38} = 87.86$, $P < .001$), with no significant changes in expectancy for the control cream ($F_{1,38} = .06$, $P = .81$) (Supplementary Fig 1).

Placebo Analgesia

The main effect of placebo analgesia (Placebo – Control) was significant ($F_{1,36} = 5.30$, $P = .027$) and marginally greater within the long conditioning group

($F_{1,36} = 4.07$, $P = .051$). Overall, placebo analgesia decreased marginally following the reveal ($F_{1,36} = 3.69$, $P = .063$). This change was not different between conditioning groups ($F_{1,36} = .20$, $P = .66$). Removing the temperature covariate from the model had no effect on the significance of these or other contrasts of placebo analgesia, discussed below.

Placebo by Reveal State

We found that the reveal resulted in increased pain under placebo conditions by 1.9 points on average ($F_{1,36} = 11.30$, $P = .002$). This increase was not significantly different between long and short participants ($F_{1,36} = .05$, $P = .83$).

Prereveal

Prereveal, placebo pain ratings were lower than control pain ratings on average ($F_{1,36} = 7.89$, $P = .008$) and were not significantly different between conditioning groups ($F_{1,36} = 1.88$, $P = .18$). However, prereveal placebo analgesia was significant in the long group ($F_{1,36} = 8.74$, $P = .005$) but not the short group ($F_{1,36} = 1.03$, $P = .32$) (Figs 3A and 3B). A post hoc analysis revealed that prereveal placebo analgesia was not correlated with the time interval between conditioning sessions in the long conditioning group ($F_{1,35} = .03$, $P = .88$).

Postreveal

Postreveal, placebo analgesia was not significant on average across both conditioning groups ($F_{1,36} = .90$, $P = .35$). However, the long group demonstrated placebo analgesia ($F_{1,36} = 4.55$, $P = .040$), and this analgesia was significantly greater than that reported by the short

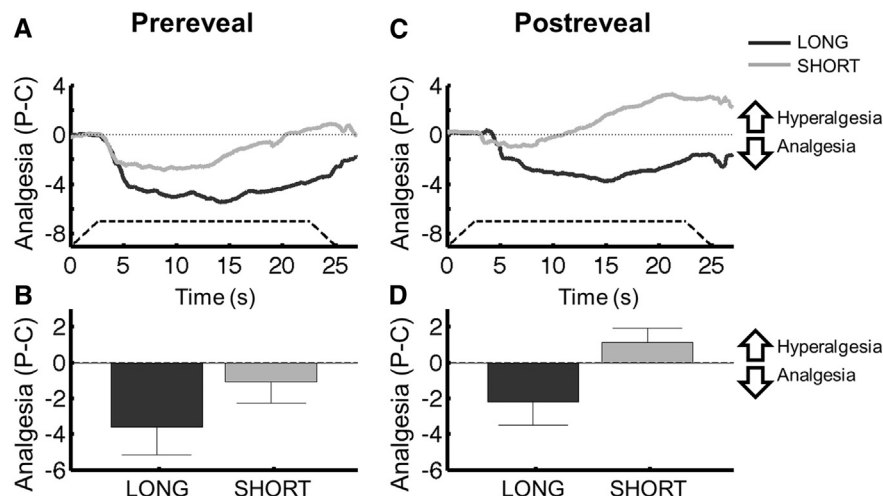


Figure 3. Placebo analgesia by conditioning group. (A) Average time course of prereveal placebo analgesia (Placebo – Control) during painful stimulation. Negative values indicate reduced pain with the placebo (analgesia), whereas positive values indicate increased pain with the placebo (hyperalgesia). The solid dark-gray and light-gray lines represent the average of subjects within the long and short conditioning groups, respectively. The dashed line shows the onset and termination of the painful stimulus and the duration of peak temperature. (B) Mean prereveal placebo analgesia. Analgesia scores are generated by averaging over the placebo effect (Placebo – Control) time course for each subject. Long subjects had significant placebo analgesia prereveal, but short subjects did not. Prereveal placebo analgesia was not different between groups. (C) Average time course of postreveal placebo analgesia during painful stimulation. The lines are defined as in panel A. (D) Mean postreveal placebo analgesia. Long subjects continued to demonstrate placebo analgesia postreveal and had significantly greater placebo analgesia than the short group. There was no significant postreveal placebo effect in the short group. Error bars represent standard errors of the mean.

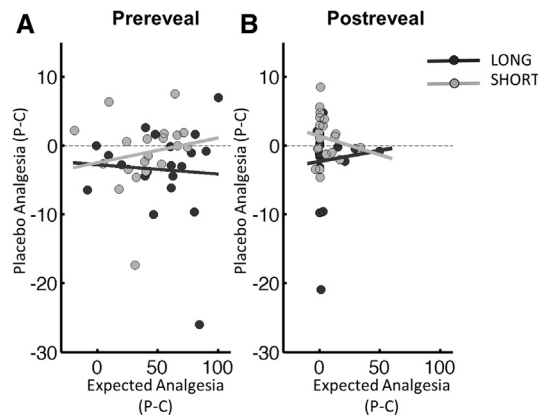


Figure 4. Placebo analgesia by expected analgesia. In both conditioning groups, placebo analgesia (Placebo – Control) was not correlated with expected analgesia (Placebo – Control) either (A) prereveal or (B) postreveal. In both panels, the dark-gray line represents the relationship between expected and reported analgesia in the long group, with the dark-gray circles representing individual subjects. Similarly, the light-gray line represents the relationship between expected and reported analgesia in the short group, with light-gray circles representing individual subjects.

group ($F_{1,36} = 4.27$, $P = .046$), who did not have a significant placebo response ($F_{1,36} = .62$, $P = .44$) (Figs 3C and 3D). A post hoc analysis revealed that postreveal placebo analgesia was not correlated with the time interval between conditioning sessions in the long conditioning group ($F_{1,35} = .15$, $P = .70$).

Placebo Analgesia by Expectancy

There was no relationship between expected analgesia and placebo analgesia either prereveal ($t_{38} = -.14$, $P = .89$) or postreveal ($t_{38} = -.04$, $P = .96$) (Fig 4).

Conditioning Order

Overall, subjects who were exposed to the control cream (and high pain) before the placebo cream (and low pain) during the initial conditioning sessions (Control first – Placebo first) demonstrated stronger placebo effects than subjects with the reversed conditioning order ($F_{1,36} = 7.10$, $P = .011$). This conditioning order effect was significantly greater prereveal compared to postreveal ($F_{1,36} = 4.78$, $P = .035$). Specifically, having a control first conditioning order was associated with stronger prereveal placebo effects on average ($F_{1,36} = 10.44$, $P = .003$) (Fig 5). Postreveal, there was no effect of conditioning order on placebo analgesia ($F_{1,36} = 1.26$, $P = .27$). These effects were not different between conditioning groups either prereveal ($F_{1,36} = .50$, $P = .48$) or postreveal ($F_{1,36} < .01$, $P = .96$).

Subjects in the control first order demonstrated significant prereveal placebo analgesia ($F_{1,36} = 18.24$, $P < .001$) that did not vary by conditioning group ($F_{1,36} = 2.16$, $P = .15$), whereas subjects in the placebo first order had no prereveal placebo response ($F_{1,36} = .09$, $P = .77$). Therefore, it was critical to conduct a secondary post hoc test to examine how placebo analgesia changed postreveal specifically among control first subjects. Postreveal, within the control first order group, long subjects continued to demonstrate placebo analgesia ($F_{1,36} = 4.19$, $P = .048$), whereas short subjects did not ($F_{1,36} < .01$, $P = .98$). There was no significant difference between these groups ($F_{1,36} = 2.04$, $P = .16$). Within the placebo first order, there was a nonsignificant trend of postreveal hyperalgesia among the short conditioning group ($F_{1,36} = 1.30$, $P = .26$) and no postreveal placebo effect in the long subjects ($F_{1,36} = .94$, $P = .34$).

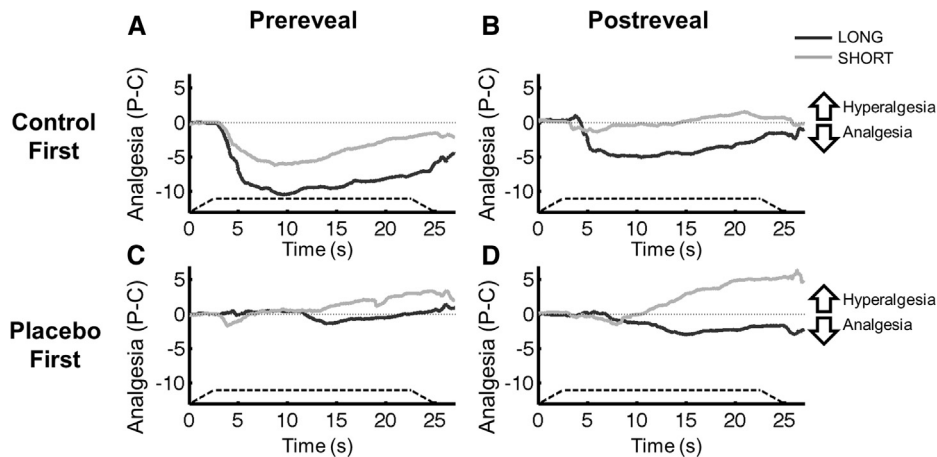


Figure 5. Placebo analgesia by conditioning order, conditioning group, and reveal state. (A) Average time course of prereveal placebo analgesia (Placebo – Control) during painful stimulation for subjects who received the control first order. Negative values indicate reduced pain with the placebo (analgesia), whereas positive values indicate increased pain with the placebo (hyperalgesia). The solid dark-gray and light-gray lines represent the average of subjects within the long and short conditioning groups, respectively. The dashed line shows the onset and termination of the painful stimulus and the duration of peak temperature. On average, subjects who received the control first order had significant placebo analgesia, and this effect was not significantly different between conditioning groups (Long – Short). (B) Average time course of postreveal placebo analgesia during painful stimulation for subjects who received the control first order. The lines are defined as in panel A. Postreveal, long subjects, but not short subjects, who received the control first order continued to demonstrate placebo analgesia. (C) Average time course of prereveal placebo analgesia during painful stimulation for subjects who received the placebo first order. The lines are defined as in panel A. Prereveal, there was no significant placebo effect for subjects who received the placebo first conditioning order. (D) Average time course of postreveal placebo analgesia during painful stimulation for subjects who received the placebo first order. The lines are defined as in panel A. Postreveal, placebo first subjects did not show significant analgesia or hyperalgesia on average.

The effect of conditioning order on placebo analgesia was not explainable in terms of habituation or other adaptation effects as in previous studies,³⁶ because participants in both order conditions received the same control–placebo–placebo–control sequence during the test session. Similarly, expectancy (Placebo – Control) was not different based on conditioning order either prereveal ($t_{38} = -1.3$, $P = .20$) or postreveal ($t_{38} = 1.0$, $P = .32$) suggesting that the observed conditioning order effects cannot be explained by differences in expected analgesia.

Discussion

This study demonstrates that multiple sessions of conditioning can lead to placebo analgesia that persists even when the true nature of a placebo treatment is convincingly revealed. Furthermore, there were no detectable differences in postreveal expected analgesia between the long and short groups, even though postreveal placebo analgesia was significantly greater in the long group. Together, these results suggest that processes not explicitly associated with reported expectancies can mediate conditioned placebo analgesia.

These results parallel emerging evidence from other studies demonstrating that placebo analgesia may occur even when subjects know they have received a placebo. In one study, placebo treatments reduced symptoms of irritable bowel syndrome and measures of clinical function even when participants were told the medication was an inert placebo.¹⁷ In another case, informing subjects that previous tests were performed using a placebo did not inhibit subsequent placebo analgesia when subjects were again told they were receiving a real analgesic.⁸ However, in both of these examples, subjects were encouraged to believe the placebo to be effective by either imagining it to assist their own healing processes or actually believing it to be an active treatment. A critical difference between the current study and previous studies is that in this study, subjects knew the treatment was a placebo, were explicitly told that it had no analgesic properties, and were not led to expect relief from a placebo treatment.

Other studies suggest that initial experiences create persistent beliefs that are subsequently hard to reverse,^{27,30} and such studies highlight the resistance of placebo effects to potentially disconfirming information. However, because most “open-label placebo” studies¹⁷ and studies demonstrating conditioned physiological effects^{1,13} have not explicitly measured belief in the placebo, these studies may demonstrate the persistence of beliefs themselves. Our results provide new information by demonstrating conditioned placebo analgesia in response to a treatment that participants believe to be ineffective.

The observed conditioning order effects are very important with respect to future studies on conditioned placebo analgesia, as the conditions under which placebo effects are maximized are not well understood.

Previous work found that eliminating the association between placebo cues and pain relief during the initial conditioning session inhibited acquisition of the placebo effect even when subsequent presentations of the placebo cues were paired with pain relief.⁹ Here, we found that even if the initial placebo cues are associated with pain relief, subsequent placebo analgesia is impaired if those cues are presented before receiving the high pain stimulations experienced during control blocks. In particular, subjects in the short conditioning group who received the placebo first order appeared to have a mild hyperalgesic response to the placebo. Although not significant, this hyperalgesia likely contributed to the finding that short subjects reported less placebo analgesia than long subjects postreveal. Critically, both long and short subjects in the control first order reported analgesia prereveal, though only the long subjects continued to experience analgesia postreveal.

The mechanisms underlying the effect of conditioning order on analgesia are unclear and deserve further investigation at both the psychological and neural levels. Acquisition of placebo analgesia may rely on a reduction in stress that favors subsequent learning effects.²⁰ The feeling of relief when transitioning from the control cream to the placebo cream could enhance placebo analgesia,¹¹ whereas the stress of transitioning from the placebo cream condition to the more painful control cream condition could inhibit acquisition of placebo analgesia²⁰ and even induce hyperalgesia.⁵ The reduced placebo analgesia in the placebo first group may also result from a relative judgment effect. Experiencing the more painful control stimuli followed by less painful placebo conditioning stimuli invites a relative comparison that focuses on pain reduction with the placebo, whereas beginning with the mildly painful placebo stimulus instead highlights the painfulness of the placebo stimulation absent the context of the more painful control stimuli. Focus on the painfulness of the placebo stimuli may inhibit acquisition of placebo analgesia similar to when the pain is not reduced for the placebo.⁹

There were several limitations to this study. Given that participants had participated in all experimental sessions with the same experimenter (S.M.S.), the experimenter was never blinded to which conditioning group subjects were in. This could be mitigated in the future by having a separate experimenter conduct the conditioning portion of the study. A second limitation is that there is no titration of expectation in our manipulation as we believed it was important for subjects to believe they had received an analgesic during conditioning given that certainty in a treatment typically leads to stronger placebo analgesia than uncertainty.²² Future studies could include subject groups who are told they “may have” or “have not” received an analgesic during conditioning sessions. There is some difficulty in interpreting the relationship between expectancy and analgesia postreveal, as expected analgesia was near zero with little variance. Finally, subjects who received the placebo first order did not acquire placebo analgesia

prereveal. Future studies aiming to condition placebo analgesia should use a control first, as opposed to placebo first, order.⁹

To our knowledge, this is the first study to demonstrate reduced pain to a conditioned stimulus in healthy subjects who are fully aware that they are receiving an inert treatment without the use of any pharmacologic agents. Here, we have demonstrated that we can use different levels of conditioning to elicit reliably different effects from an expectation reversal, and demonstrate that conditioning can suppress the effects of a reversal of beliefs even when no difference in expectations is observed between conditioning groups. This has several implications for medical contexts. We speculate that some variant of the design used in this study may be used to wean patients with acute pain (eg, postoperative pain) off painkillers in a way that could lower the potential for future addiction

by substituting a placebo for a chemically active treatment while simultaneously keeping pain reduced. We suspect that the repeated administration of a (potentially nonopioid) analgesic² combined with a subsequent enhancement of expected analgesia from a placebo treatment¹⁷ would yield the strongest placebo effects. In addition to differences in analgesia by conditioning group, differences by conditioning order may explain why people experience either relief or symptom worsening from a variety of medical treatments based on differences in beliefs and prior learning, highlighting the importance of psychology in medicine.

Supplementary Data

Supplementary data related to this article can be found online at <http://dx.doi.org/10.1016/j.jpain.2014.12.008>.

References

1. Albring A, Wendt L, Benson S, Witzke O, Kribben A, Engler H, Schedlowski M: Placebo effects on the immune response in humans: The role of learning and expectation. *PLoS One* 7:e49477, 2012
2. Amanzio M, Benedetti F: Neuropharmacological dissection of placebo analgesia: expectation-activated opioid systems versus conditioning-activated specific subsystems. *J Neurosci* 19:484-494, 1999
3. Au Yeung ST, Colagiuri B, Lovibond PF, Colloca L: Partial reinforcement, extinction, and placebo analgesia. *Pain* 155:1110-1117, 2014
4. Bates D, Maechler M, Bolker B, Walker S: lme4: Linear mixed-effects models using Eigen and S4. R package version 1.0-6; 2014
5. Benedetti F, Amanzio M, Vighetti S, Asteggiano G: The biochemical and neuroendocrine bases of the hyperalgesic nocebo effect. *J Neurosci* 26:12014-12022, 2006
6. Benedetti F, Pollo A, Lopiano L, Lanotte M, Vighetti S, Rainero I: Conscious expectation and unconscious conditioning in analgesic, motor, and hormonal placebo/nocebo responses. *J Neurosci* 23:4315-4323, 2003
7. Bingel U, Wanigasekera V, Wiech K, Ni Mhuircheartaigh R, Lee MC, Ploner M, Tracey I: The effect of treatment expectation on drug efficacy: Imaging the analgesic benefit of the opioid remifentanyl. *Sci Transl Med* 3:70ra14, 2011
8. Chung SK, Price DD, Verne GN, Robinson ME: Revelation of a personal placebo response: Its effects on mood, attitudes and future placebo responding. *Pain* 132:281-288, 2007
9. Colloca L, Benedetti F: How prior experience shapes placebo analgesia. *Pain* 124:126-133, 2006
10. Colloca L, Petrovic P, Wager TD, Ingvar M, Benedetti F: How the number of learning trials affects placebo and nocebo responses. *Pain* 151:430-439, 2010
11. Flaten MA, Aslaksen PM, Lyby PS, Bjorkedal E: The relation of emotions to placebo responses. *Philos Trans R Soc Lond B* 366:1818-1827, 2011
12. Gallistel CR, Fairhurst S, Balsam P: The learning curve: Implications of a quantitative analysis. *Proc Natl Acad Sci U S A* 101:13124-13131, 2004
13. Goebel MU, Trebst AE, Steiner J, Xie YF, Exton MS, Frede S, Canbay AE, Michel MC, Heemann U, Schedlowski M: Behavioral conditioning of immunosuppression is possible in humans. *FASEB J* 16:1869-1873, 2002
14. Goffaux P, Redmond WJ, Rainville P, Marchand S: Descending analgesia—When the spine echoes what the brain expects. *Pain* 130:137-143, 2007
15. Jensen KB, Kaptchuk TJ, Kirsch I, Raicek J, Lindstrom KM, Berna C, Gollub RL, Ingvar M, Kong J: Nonconscious activation of placebo and nocebo pain responses. *Proc Natl Acad Sci U S A* 109:15959-15964, 2012
16. Jepma M, Jones M, Wager TD: The dynamics of pain: Evidence for simultaneous site-specific habituation and site-nonspecific sensitization in thermal pain. *Pain* 15: 734-746, 2014
17. Kaptchuk TJ, Friedlander E, Kelley JM, Sanchez MN, Kokkotou E, Singer JP, Kowalczykowski M, Miller FG, Kirsch I, Lembo AJ: Placebos without deception: A randomized controlled trial in irritable bowel syndrome. *PLoS One* 5:e15591, 2010
18. Kirsch I, Kong J, Sadler P, Spaeth R, Cook A, Kaptchuk TJ, Gollub R: Expectancy and conditioning in placebo analgesia: Separate or connected processes? *Psychol Conscious* 1:51-59, 2014
19. Kirsch I, Lynn SJ, Vigorito M, Miller RR: The role of cognition in classical and operant conditioning. *J Clin Psychol* 60:369-392, 2004
20. Lyby PS, Aslaksen PM, Flaten MA: Is fear of pain related to placebo analgesia? *J Psychosom Res* 68:369-377, 2010
21. Montgomery GH, Kirsch I: Classical conditioning and the placebo effect. *Pain* 72:107-113, 1997
22. Pollo A, Amanzio M, Arslanian A, Casadio C, Giuliano M, Benedetti F: Response expectancies in placebo analgesia and their clinical relevance. *Pain* 93:77-84, 2001
23. Price DD, Craggs J, Verne GN, Perlstein WM, Robinson ME: Placebo analgesia is accompanied by large

reductions in pain-related brain activity in irritable bowel syndrome patients. *Pain* 127:63-72, 2007

24. Price DD, Finniss DG, Benedetti F: A comprehensive review of the placebo effect: Recent advances and current thought. *Annu Rev Psychol* 59:565-590, 2008

25. Price DD, Milling LS, Kirsch I, Duff A, Montgomery GH, Nicholls SS: An analysis of factors that contribute to the magnitude of placebo analgesia in an experimental paradigm. *Pain* 83:147-156, 1999

26. Rescorla RA, Wagner AR: A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In: Black AH, Prokasy WF (eds): *Classical Conditioning II: Current Research and Theory*. Appleton-Century-Crofts, 1972, pp 64-99

27. Ross L, Lepper MR, Hubbard M: Perseverance in self-perception and social perception: Biased attributional processes in the debriefing paradigm. *J Pers Soc Psychol* 32:880-892, 1975

28. Sandler AD, Bodfish JW: Open-label use of placebos in the treatment of ADHD: A pilot study. *Child* 34:104-110, 2008

29. Sandler AD, Glesne CE, Bodfish JW: Conditioned placebo dose reduction: A new treatment in attention-deficit hyperactivity disorder? *J Dev Behav Pediatr* 31:369-375, 2010

30. Staudinger MR, Buchel C: How initial confirmatory experience potentiates the detrimental influence of bad advice. *NeuroImage* 76:125-133, 2013

31. Stewart-Williams S, Podd J: The placebo effect: Dissolving the expectancy versus conditioning debate. *Psychol Bull* 130:324-340, 2004

32. Vase L, Petersen GL, Riley JL 3rd, Price DD: Factors contributing to large analgesic effects in placebo mechanism studies conducted between 2002 and 2007. *Pain* 145:36-44, 2009

33. Voudouris NJ, Peck CL, Coleman G: Conditioned placebo responses. *J Pers Soc Psychol* 48:47-53, 1985

34. Voudouris NJ, Peck CL, Coleman G: The role of conditioning and verbal expectancy in the placebo response. *Pain* 43:121-128, 1990

35. Wager TD, Fields HL: Placebo analgesia. In: Wall PD, Melzack R (eds): *Textbook of Pain*; 2013, pp 362-373

36. Wager TD, Matre D, Casey KL: Placebo effects in laser-evoked pain potentials. *Brain Behav Immun* 20: 219-230, 2006

37. Watson A, El-Deredy W, Bentley DE, Vogt BA, Jones AK: Categories of placebo response in the absence of site-specific expectation of analgesia. *Pain* 126:115-122, 2006

38. Watson A, El-Deredy W, Iannetti GD, Lloyd D, Tracey I, Vogt BA, Nadeau V, Jones AK: Placebo conditioning and placebo analgesia modulate a common brain network during pain anticipation and perception. *Pain* 145:24-30, 2009